

Assist Home Training Table Tennis Skill Acquisition via Immersive Learning and Web Technologies

Jian-Jia Weng

Yu-Hsin Wang

Calvin Ku*

Dong-Xian Wu

Yi-Min Lau

Wan-Lun Tsai

Tse-Yu Pan

Min-Chun Hu†

Hung-Kuo Chu

Te-Cheng Wu

National Tsing Hua University, Taiwan

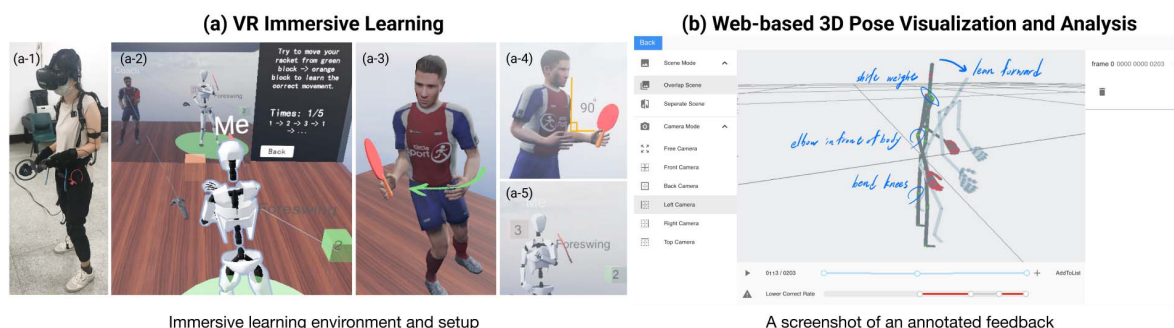


Figure 1: Illustration of our VR learning and web-based feedback system. (a-1) Participant wearing IMU suit for motion capture (a-2) An overview of our immersive training environment from a third person viewing perspective. (a-3) An example of key concept visualization of the body rotation for backswing (a-4) An example of key concept visualization for demonstrating the angle between the upper arm and the forearm (a-5) Virtual boxes to guide users about forehand drive racket position. (b) Our web-based 3D pose feedback application.

ABSTRACT

Sports applications in Virtual Reality (VR) have become immensely popular for training skill-based sports like table tennis. However, the existing researches do not focus on designing an intuitive system for efficient communication between the trainee and the coach. We developed a VR table tennis training system for table tennis skill acquisition that focuses on helping coaches to convey a trainee's mistake clearly. Our system consists of a VR training system where trainees can learn a skill gradually and a web-based feedback annotative tool for coaches. Trainees can examine their mistakes through a tablet or an immersive VR world.

Index Terms: Human-centered computing—Visualization—Visualization systems and tools—Visualization toolkits; Human-centered computing—Interaction design—Interaction design process and methods—Activity centered design

1 INTRODUCTION

Digitalization and big data have brought an unprecedented impact on the development of the sports industry. Traditionally, data collection in sports requires extensive paperwork and post-practice efforts from the training staff. While the athlete practices, the training staff has to take physical notes and convert the accumulated notes into charts or graphs to represent an athlete's performance. Nowadays, individual sports data can be recorded using wearable monitors and accessible by the internet using mobile devices. In addition, due to the COVID-19 pandemic, many online fitness clubs and influencers use social media platforms to engage in fitness activities anywhere

and anytime with their followers and members. Now exercising at home has become a new trend. In this work, we aim to study the feasibility of home training based on VR and web technology.

Existing table tennis games [2] focus on entertainment aspects, and some researches [3–5] have explored different ways for table tennis skill acquisitions. However, after analyzing the learning dilemmas of trainees from different perspectives via conducting semi-structured user interviews with two coaches and six senior players in a varsity table tennis team, we found that the existing systems fail to minimize the communication gap between the coach and the trainee while adjusting the trainee's postures. The trainees and coaches need a more intuitive tool to have two-way communication while correcting the movements, especially for remote communication. In this work, we develop a system for beginners to learn and obtain clear feedback from coaches virtually. We also proved that trainees could transfer the movements learned in our system (consisting of a VR training application and a web-based feedback annotative tool) to real-world training.

2 FRAMEWORK

Our system consists of a virtual reality application developed with Unity and a mobile application developed with ReactJS. Besides that, we utilize a Perception Neuron IMU suit for 3D human pose capture. As shown in Figure 2, these subsystems communicate with each other via a Hypertext Transfer Protocol (HTTP) server. Our backend server is developed with NodeJS to retrieve and update data from a MongoDB data storage system. Since correct and good postures can stabilize a trainee's performance, our system uses an IMU-based motion capture suit (Perception Neuron) to record the professional coach's pose of forehand drive in table tennis. The user (coach or trainee) can view the recorded coach's pose in VR from different perspectives. In addition, users can record and upload their poses to the server, and use the web feedback tool to visualize, annotate, and share the movement learning results.

*Corresponding Author 1: calvinku1209@gmail.com

†Corresponding Author 2: anitahu@cs.nthu.edu.tw

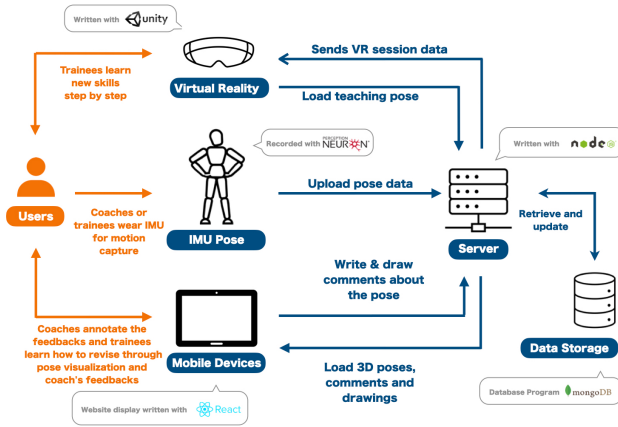


Figure 2: System framework.

2.1 Training in VR immersive environment

Figure 1(a) shows our immersive learning environment, in which step by step learning instructions are given with intuitive visual hints (e.g. numbers indicating moving order and arrows indicating moving trajectories). An animation of the coach's skeleton is presented on the user's left-hand side. In addition, a looped mnemonic audio phrase about the key concepts of the swing is played repeatedly so users can remember the steps more easily. Our system also contains a mirror in the virtual scene so the user can see front view of the coach's posture and their own posture (collected by an IMU suit) clearly when performing the action. Unlike conventional pose analysis tools, which only compare joint's position/angle distance, we further track the trainee's center of gravity when they are performing the skills. That is because correct change in the center of gravity helps to perform the skill more smoothly and increase stability of successful ball return. After learning the concept of a new skill, the user is asked to repeatedly practice what they had learned for five times, and our system validates if they had done so correctly based on the joint positions and the center of gravity.

2.2 Web-based 3D pose feedback

Figure 1(b) shows the interface page of our web-based visualization and feedback system. The user can view their movements with a playback slider in the bottom of the page. In the middle of the page is a 3D canvas, where users can view the poses at a specific time from different perspectives by zooming in/out, rotating and translating the virtual camera. The canvas contains the target pose displayed in light grey color and the user's pose in dark grey color. The body joints are shown as circles, and their corresponding colors represent the correctness in comparison to the target pose (green is correct and red is wrong). The similarity of two pose sequences is calculated by dynamic time warping (DTW) algorithm based on the corresponding joints' euclidean distance. Beneath the playback slider is a pose comparison indicator that highlights time intervals with low correct rate. The indicator helps the user quickly identify the frames with problematic poses. By dragging the handles in the playback slider, the user can specify an interval that he/she wants to analyze and draw/annotate information on the canvas frames. Trainers can use audio function to explain the problem of the poses.

3 EXPERIMENTS

To evaluate the effectiveness of using the proposed system to learn a new skill, we adopted a within-subject experiment. Since our target users are table tennis beginners, we invited eight participants

(4 males and 4 females) who are right-handed, never received any formal table tennis training, and aged between 21 and 27. The participants were split into two groups, and each participant experienced two training sessions while wearing an IMU suit. The first group received physical training with physical feedback from a professional table tennis coach and then used our system with remote feedback for training in the second session. The second group does the opposite (VR with remote feedback then physical training with physical feedback). After each training session, the participant was asked to physically return 15 balls served from a table tennis serve machine. We propose two different measurements to evaluate participants' learning performance. The first one is Mean Per Joint Error (MPJPE) [1], and the other is the number of successfully returned balls. The results in MPJPE reveal that physical training with physical feedback and then using our system can help improve the trainee's form by 287.69mm. If the trainee uses our system alone, their poses can improve by 295.21mm and further improve by an extra 172.74mm after physical training. We discovered that the number of successful return did not have significant difference for each training session of both groups. The reason might be the participant's racket control is unstable when they try to adjust their posture within a short training period for both training methods.

Based on our final user interview, our participants found our step-by-step visualizing instructions can enhance the participants' understanding of a new skill. The visual cues with numbers and arrows are helpful to understand where to swing in the real world. Participants also found pose visualization is helpful to notice details that are not easy to discover in the real world. They could adjust their postures immediately after watching the virtual coach's movement by looking at the VR environment's mirror. The proposed web-based feedback system helped one participant notice that his left arm was not bent and another participant realize his feet were too stiff. After adjusting poses according to our web-based feedback application, they felt their movements became more stable and their body swing became smoother while hitting the ball. Additionally, the coach shared that our web-based feedback application allows him to observe the trainee's overall body movement and center of gravity more clearly via free viewing perspectives, which consequently helped him find posture problems efficiently.

4 CONCLUSION

In this paper, we propose a VR table tennis training system with a web-based feedback application for trainees to acquire new skills at home and receive clear feedback from a remote coach. We conducted both quantitative and qualitative evaluation of the proposed system, and the experimental results demonstrate that the trainees could transfer the movements learned in our system to real-world training.

ACKNOWLEDGMENTS

This research was supported by the Ministry of Science and Technology (Contract MOST 108-2221-E-007-106-MY3 and 109-2221-E-007-095-MY3), Taiwan.

REFERENCES

- [1] C. Ionescu, D. Papava, V. Olaru, and C. Sminchisescu. Human3. 6m: Large scale datasets and predictive methods for 3d human sensing in natural environments. *IEEE transactions on pattern analysis and machine intelligence*, 36(7):1325–1339, 2013.
- [2] F. F. Labs. Eleven table tennis.
- [3] H. Oagaz, B. Schoun, and M.-H. Choi. Performance improvement and skill transfer in table tennis through training in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 2021.
- [4] S. R. M. Systems. Table tennis biomechanics.
- [5] E. Wu, M. Piekenbrock, T. Nakamura, and H. Koike. Spinpong-virtual reality table tennis skill acquisition using visual, haptic and temporal cues. *IEEE Transactions on Visualization and Computer Graphics*, 27(5):2566–2576, 2021.